

Day-8

1. create a vehicle class, derive car and Bike classes

using system;

class vehicle

{

public void start()

{

Console.WriteLine("vehicle started");

}

}

class car : vehicle

{

public void showCar()

{

Console.WriteLine("This is a car");

}

}

class Bike : vehicle

{

public void showBike()

{

Console.WriteLine("This is a Bike");

}

}

class program

{


```
static void main ()
```

```
{
```

```
Car car = new Car();
```

```
car.start();
```

```
car.showCar();
```

```
Bike bike = new Bike();
```

```
bike.start();
```

```
bike.showBike();
```

```
}
```

```
}
```

2) Designing a payment system (polymorphism)

using System

```
class Payment
```

```
{
```

```
public virtual void makePayment()
```

```
{
```

```
Console.WriteLine("making payment");
```

```
}
```

```
}
```



```
class credit card : payment
```

```
{
```

```
public override void make payment ()
```

```
{
```

```
console.WriteLine ("payment using credit card");
```

```
}
```

```
}
```

```
class paypal : payment
```

```
{
```

```
public override void make payment ()
```

```
{
```

```
console.WriteLine ("payment using UPI paypal");
```

```
}
```

```
}
```

```
class UPI : payment
```

```
{
```

```
public override void make payment ()
```

```
{
```

```
console.WriteLine ("payment using UPI");
```

```
}
```

```
}
```

class program

{

static void main ()

{

Payment P ~~new Payment()~~

P = new credit card ();

P. Make Payment ();

P = new paypal ();

P. Make Payment ();

P = new UPI ();

P. Make Payment ();

}

}

3. implement operator overloading

using system

class number

```
{  
    public int value;
```

```
    public Number (int value)
```

```
{  
        Value = value;
```

```
}
```

```
    public static Number operator +(Number n1, Number n2)
```

```
{  
        return new Number (n1.value + n2.value);
```

```
}
```

```
}
```

class program

```
{  
    static void main ()
```

```
{  
        Number num1 = new Number (10);
```

```
        Number num2 = new Number (20);
```

```
        Number result = num1 + num2;
```

```
        Console.WriteLine("sum = " + result.value);
```

```
}
```

```
}
```